

Name Karthik

Roll no :- 261100690024

① Encrypt "Chase Your Dreams" using :

a. Caesar Cipher

b. Shift cipher with key 7

a) $C = E(P+3) \text{ mod } 26$ C. ROT 13

$$C \rightarrow (P+3) \text{ mod } 26 = (2+3) \text{ mod } 26 = 5 \text{ mod } 26 = 5 = F$$

$$H \rightarrow (P+3) \text{ mod } 26 = (7+3) \text{ mod } 26 = 10 = K$$

$$A \rightarrow (P+3) \text{ mod } 26 = (0+3) \text{ mod } 26 = 3 = D$$

$$S \rightarrow (P+3) \text{ mod } 26 = (18+3) \text{ mod } 26 = 21 = V$$

$$E \rightarrow (P+3) \text{ mod } 26 = (4+3) \text{ mod } 26 = 7 = H$$

$$Y \rightarrow (P+3) \text{ mod } 26 = (24+3) \text{ mod } 26 = 1 = B$$

$$O \rightarrow (P+3) \text{ mod } 26 = (14+3) \text{ mod } 26 = 17 = R$$

$$U \rightarrow (P+3) \text{ mod } 26 = (20+3) \text{ mod } 26 = 23 = X$$

$$R \rightarrow (P+3) \text{ mod } 26 = (17+3) \text{ mod } 26 = 20 = U$$

$$D \rightarrow (P+3) \text{ mod } 26 = (3+3) \text{ mod } 26 = 6 = G$$

$$R \rightarrow (P+3) \text{ mod } 26 = (17+3) \text{ mod } 26 = 20 = U$$

$$E \rightarrow (P+3) \text{ mod } 26 = (4+3) \text{ mod } 26 = 7 = H$$

$$A \rightarrow (P+3) \text{ mod } 26 = (0+3) \text{ mod } 26 = 3 = D$$

$$M \rightarrow (P+3) \text{ mod } 26 = (12+3) \text{ mod } 26 = 15 = P$$

$$S \rightarrow (P+3) \text{ mod } 26 = (18+3) \text{ mod } 26 = 21 = V$$

Cypher Txt = FKDVH BKXU GUHDPV

b) Shift Cypher : Key = 7

$$\text{Formula } C = E(P+7) \bmod 26$$

$$\begin{aligned} C &\rightarrow (P+7) \bmod 26 = (2+7) \bmod 26 = 9 = J \\ H &\rightarrow (P+7) \bmod 26 = (7+7) \bmod 26 = 14 = O \\ A &\rightarrow (P+7) \bmod 26 = (0+7) \bmod 26 = 7 = H \\ S &\rightarrow (P+7) \bmod 26 = (18+7) \bmod 26 = 25 = Z \\ E &\rightarrow (P+7) \bmod 26 = (4+7) \bmod 26 = 11 = L \\ Y &\rightarrow (P+7) \bmod 26 = (24+7) \bmod 26 = 5 = F \\ O &\rightarrow (P+7) \bmod 26 = (14+7) \bmod 26 = 21 = V \\ U &\rightarrow (P+7) \bmod 26 = (20+7) \bmod 26 = 1 = B \\ R &\rightarrow (P+7) \bmod 26 = (17+7) \bmod 26 = 24 = Y \\ D &\rightarrow (P+7) \bmod 26 = (3+7) \bmod 26 = 10 = K \\ K &\rightarrow (P+7) \bmod 26 = (17+7) \bmod 26 = 24 = Y \\ E &\rightarrow (P+7) \bmod 26 = (4+7) \bmod 26 = 11 = L \\ A &\rightarrow (P+7) \bmod 26 = (0+7) \bmod 26 = 7 = H \\ M &\rightarrow (P+7) \bmod 26 = (12+7) \bmod 26 = 19 = T \\ S &\rightarrow (P+7) \bmod 26 = (18+7) \bmod 26 = 25 = Z \end{aligned}$$

Cypher Txt: JOH2L FVB4 K4LHTZ

c) ROT 13 $\rightarrow C = E(P+13) \bmod 26$

$$\begin{aligned} C &\rightarrow (P+13) \bmod 26 = (2+13) \bmod 26 = P \\ H &\rightarrow (P+13) \bmod 26 = (7+13) \bmod 26 = U \\ A &\rightarrow (P+13) \bmod 26 = (0+13) \bmod 26 = N \\ S &\rightarrow (P+13) \bmod 26 = (18+13) \bmod 26 = F \\ E &\rightarrow (P+13) \bmod 26 = (4+13) \bmod 26 = R \\ Y &\rightarrow (P+13) \bmod 26 = (24+13) \bmod 26 = L \\ O &\rightarrow (P+13) \bmod 26 = (14+13) \bmod 26 = B \\ U &\rightarrow (P+13) \bmod 26 = (20+13) \bmod 26 = I \\ R &\rightarrow (P+13) \bmod 26 = (17+13) \bmod 26 = E \\ D &\rightarrow (P+13) \bmod 26 = (3+13) \bmod 26 = Q \\ K &\rightarrow (P+13) \bmod 26 = (17+13) \bmod 26 = E \\ E &\rightarrow (P+13) \bmod 26 = (4+13) \bmod 26 = R \\ A &\rightarrow (P+13) \bmod 26 = (0+13) \bmod 26 = N \\ M &\rightarrow (P+13) \bmod 26 = (12+13) \bmod 26 = Z \\ S &\rightarrow (P+13) \bmod 26 = (18+13) \bmod 26 = F \end{aligned}$$

CTJ

PUNER LBHE QERNZF

②

a. "WRWDOOB VH FWH PHVVDJH" using Caesar Cypher

$$\Rightarrow P = D(C-3) \bmod 26$$

$$\begin{aligned} W &\rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T \\ R &\rightarrow (C-3) \bmod 26 = (17-3) \bmod 26 = 14 = O \\ W &\rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T \\ D &\rightarrow (C-3) \bmod 26 = (3-3) \bmod 26 = 0 = A \\ O &\rightarrow (C-3) \bmod 26 = (14-3) \bmod 26 = 11 = L \\ O &\rightarrow (C-3) \bmod 26 = (14-3) \bmod 26 = 11 = L \\ B &\rightarrow (C-3) \bmod 26 = (1-3) \bmod 26 = 24 = Y \\ V &\rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S \\ H &\rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E \\ F &\rightarrow (C-3) \bmod 26 = (5-3) \bmod 26 = 2 = C \\ U &\rightarrow (C-3) \bmod 26 = (20-3) \bmod 26 = 17 = R \\ H &\rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E \\ W &\rightarrow (C-3) \bmod 26 = (22-3) \bmod 26 = 19 = T \\ P &\rightarrow (C-3) \bmod 26 = (15-3) \bmod 26 = 12 = M \\ H &\rightarrow (C-3) \bmod 26 = (7-3) \bmod 26 = 4 = E \\ V &\rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S \\ V &\rightarrow (C-3) \bmod 26 = (21-3) \bmod 26 = 18 = S \\ D &\rightarrow (C-3) \bmod 26 = (3-3) \bmod 26 = 0 = A \\ J &\rightarrow (C-3) \bmod 26 = (9-3) \bmod 26 = 6 = G \\ H &\rightarrow (C-3) \bmod 26 = (9-3) \bmod 26 = 6 = G \end{aligned}$$

PT = Totally Secret message

b. "STB DTWP YNRS" Shift Cypher, 5

$$P = D(C-5) \bmod 26$$

$$\begin{aligned} S &\rightarrow (C-5) \bmod 26 = (18-5) \bmod 26 = 13 = N \\ T &\rightarrow (C-5) \bmod 26 = (9-5) \bmod 26 = 4 = E \\ B &\rightarrow (C-5) \bmod 26 = (1-5) \bmod 26 = 22 = W \\ D &\rightarrow (C-5) \bmod 26 = (3-5) \bmod 26 = 24 = Y \\ T &\rightarrow (C-5) \bmod 26 = (19-5) \bmod 26 = 14 = O \\ W &\rightarrow (C-5) \bmod 26 = (22-5) \bmod 26 = 17 = R \\ P &\rightarrow (C-5) \bmod 26 = (15-5) \bmod 26 = 10 = K \\ Y &\rightarrow (C-5) \bmod 26 = (24-5) \bmod 26 = 19 = T \\ N &\rightarrow (C-5) \bmod 26 = (13-5) \bmod 26 = 8 = H \\ R &\rightarrow (C-5) \bmod 26 = (17-5) \bmod 26 = 12 = M \\ J &\rightarrow (C-5) \bmod 26 = (9-5) \bmod 26 = 4 = E \end{aligned}$$

2F

c) "FRAQ ZBER PBSSRK" using ROT 13

$$P = D(c+13) \bmod 26$$

$$F \rightarrow (c+13) \bmod 26 = (5+13) \bmod 26 = S$$

$$R \rightarrow (c+13) \bmod 26 = (17+13) \bmod 26 = E$$

$$A \rightarrow (c+13) \bmod 26 = (0+13) \bmod 26 = N$$

$$Q \rightarrow (c+13) \bmod 26 = (16+13) \bmod 26 = D$$

$$Z \rightarrow (c+13) \bmod 26 = (25+13) \bmod 26 = M$$

$$B \rightarrow (c+13) \bmod 26 = (1+13) \bmod 26 = O$$

$$E \rightarrow (c+13) \bmod 26 = (4+13) \bmod 26 = K$$

$$R \rightarrow (c+13) \bmod 26 = (17+13) \bmod 26 = E$$

$$P \rightarrow (c+13) \bmod 26 = (15+13) \bmod 26 = C$$

$$B \rightarrow (c+13) \bmod 26 = (1+13) \bmod 26 = O$$

$$S \rightarrow (c+13) \bmod 26 = (18+13) \bmod 26 = F$$

$$S \rightarrow (c+13) \bmod 26 = (18+13) \bmod 26 = F$$

$$R \rightarrow (c+13) \bmod 26 = (17+13) \bmod 26 = E$$

$$R \rightarrow (c+13) \bmod 26 = (17+13) \bmod 26 = E$$

PT = SEND MORE COFFEE